ECON 100B Spring 2026 Your name and SID: Marcus Huang 3041432436

Problem Set 1

Due on Gradescope: check the course Drive calendar
Consult the Late Policy in the syllabus

Instructions: Work with this file as a template, perhaps by File: Download: Microsoft Word. Other options, like handwriting, are fine as long as they are legible. Name and SID at the top.

Did you work with other students? Identify them below. **If you work with others, you should write answers in your own words.** (If you understand your answers, this should be easy.)

Type your name to sign the Honor Code. Save as a PDF, upload to Gradescope, and click through to locate each question part before clicking Submit. Profit.

You will see point totals associated with each question. But we will score your problem sets based on completion, not correctness. The point totals are there to make you familiar with how the exams will work, which are scored on correctness.

Questions below:

Did you work with other students? List them below:

No, I'm finishing this Problem set on my own.

Please type your name as an affirmation of the Honor Code at the University of California, Berkeley.

> "As a member of the UC Berkeley community, I act with honesty, integrity, and respect for others."

Type your name:

Marcus Huang

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1. [14 points total] Short answers with capital answers.

(a) [2 points] Recall that GDP measured on the expenditure side is given by

$$Y = C + I + G + NX \tag{1.1}$$

Roughly how big are each of these components as a *percentage* of GDP in the U.S. today?

Approximately : $C \approx 62\%$ $I \approx 22\%$ $G \approx 21\%$

$NX \approx -5\%$

(b) [2 points] What exactly does investment spending create? Explain.

I

Investment is spending on creating new capital goods, which we will use in the future to produce other products.
For example, the factories, office buildings, software, computers.
We spend money on it then use it to earn more benifits.

(c) [2 points] In the first half of ECON 100B, what is capital? Describe it in a few sentences. Maybe describe the economic meanings of the word that do NOT directly apply to our study of economic growth.

Instead of money or cash. capital in Econ 100B is more refer to physical stuff like machines and equipments, those can be used to produce new products.

(d) [2 points] Recall that real GDP Y^R equals nominal GDP Y^N divided by the price level:

$$Y^R = \frac{Y^N}{P} \quad (1.2)$$

price

Take the growth rate of both sides of equation (1.2), and recall that the growth rate of the price level is inflation and denoted by a lowercase Greek letter "pi," π. If the growth rate in nominal GDP is 5% and inflation is 2%, what is the growth rate of real GDP?

$Y^R = \frac{Y^N}{P} \rightarrow g(Y^R) = g(Y^N) - g(P) \rightarrow \pi = g(P)$

$\Rightarrow g(Y^R) = g(Y^N) - \pi \rightarrow g(Y^R) = 5\% - 2\% = 3\%$

(e) [2 points] In Chapter 4, we will see that GDP is modeled well by this Cobb-Douglas production function:

$$Y = A K^{\alpha} L^{1-\alpha} \quad (1.3)$$

where α is a constant number between 0 and 1 that is also capital's share of income. In the textbook, $\alpha = 1/3$, which is an appropriate level of the parameter for U.S. data.

Take the growth rate of both sides of equation (1.3), treating A, K, and L as variables that potentially have growth rates. Simplify and briefly describe your answer:

$Y = AK^{\alpha}L^{1-\alpha}$, $\alpha = 1/3 \rightarrow g(Y) = g(A) + \alpha g(K) + (1-\alpha) g(L)$

$\Rightarrow g(Y) = g(A) + \frac{1}{3} g(K) + \frac{2}{3} g(L)$

⇒ with $\alpha = 1/3$, the growth rate of output is equals to the sum of the productivity growth rate, 1/3 growth of capital and $\frac{2}{3}$ growth of labor.

(f) [2 points] Using your answer above, describe what happens to Y if A is constant while K and L both increase by 2%. Be specific and briefly explain.

A is constant $\rightarrow g(A) = 0$ then $g(K) = g(L) = 2\%$

thus $g(A) + \frac{1}{3}g(K) + \frac{2}{3}g(L) = 0 + \frac{2}{3}\% + \frac{4}{3}\% = \frac{6}{3}\% = 2\%$

$\Rightarrow g(Y) = 2\%$

⇒ the output Y was increase by 2% as well.

(g) [2 points] Using your answer above, describe what happens to Y if A grows by 3% while K and L both increase by 2%. Be specific and briefly explain.

Same : $g(A) = 3\%$, $g(K) = 2\%$, $g(L) = 2\%$ $\rightarrow$ $3\% + \frac{2}{3}\% + \frac{4}{3}\% = 3\% + 2\% = 5\%$

$\Rightarrow g(Y) = 5\%$

$\Rightarrow$ Y grows by 5% under this scenario

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2. [12 points total] Cobb-Douglas Come Out to Play?

Although there are some phenomena in economics that the Cobb-Douglas function cannot capture realistically, it is a remarkably versatile tool that you will likely see again and again in your undergraduate studies. Recall that in Chapter 4, we see that the U.S. macroeconomy can be modeled well using Cobb-Douglas production:

$$\boxed{Y = A K^{\alpha} L^{1-\alpha}} \tag{2.1}$$

where $\alpha = 1/3$ is capital's share of income. When we take first partial derivatives of equation (2.1) we reveal the useful functional forms of the marginal products of capital (MPK) and of labor (MPL), which in competitive markets equal their factor prices: the interest rate r and the wage w:

$\frac{\partial Y}{\partial K} = MPK = r$	$r = \frac{1}{3} A \left(\frac{L}{K}\right)^{2/3}$	(2.2)
$\frac{\partial Y}{\partial L} = MPL = w$	$w = \frac{2}{3} A \left(\frac{K}{L}\right)^{1/3}$	(2.3)

(a) [2 points] Take the growth rate of both sides of equation (2.2), which shows r as a function of A, K, and L, and simplify. If A is constant, then what must be true about the growth rate of r when labor and capital are growing at the same rate and thus the capital-labor ratio is constant? Show your work.

$r = 1/3 A \left(\frac{L}{K}\right)^{2/3} \rightarrow g(r) = g(A) + \frac{2}{3}(g(L) - g(K))$

$\Rightarrow$ If A is constant then $g(L) = g(K)$ $\rightarrow g(r) = 0 + \frac{2}{3} \cdot 0 = 0$

(b) [2 points] Using your answer above, what happens to r if there is no growth in A but K rises above L by 1%?

$g(A) = 0$, $g(K) = g(L) + 1\% \rightarrow g(L) - g(K) = g(L) - g(L) - 1\% = -1\%$

$\Rightarrow g(r) = 0 + \frac{2}{3} \cdot -1\% \rightarrow g(r) = -\frac{2}{3}\% \approx -0.67\%$

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Consider what happens when we take the natural log of both sides of equation (2.3), which is similar to taking the growth rate:

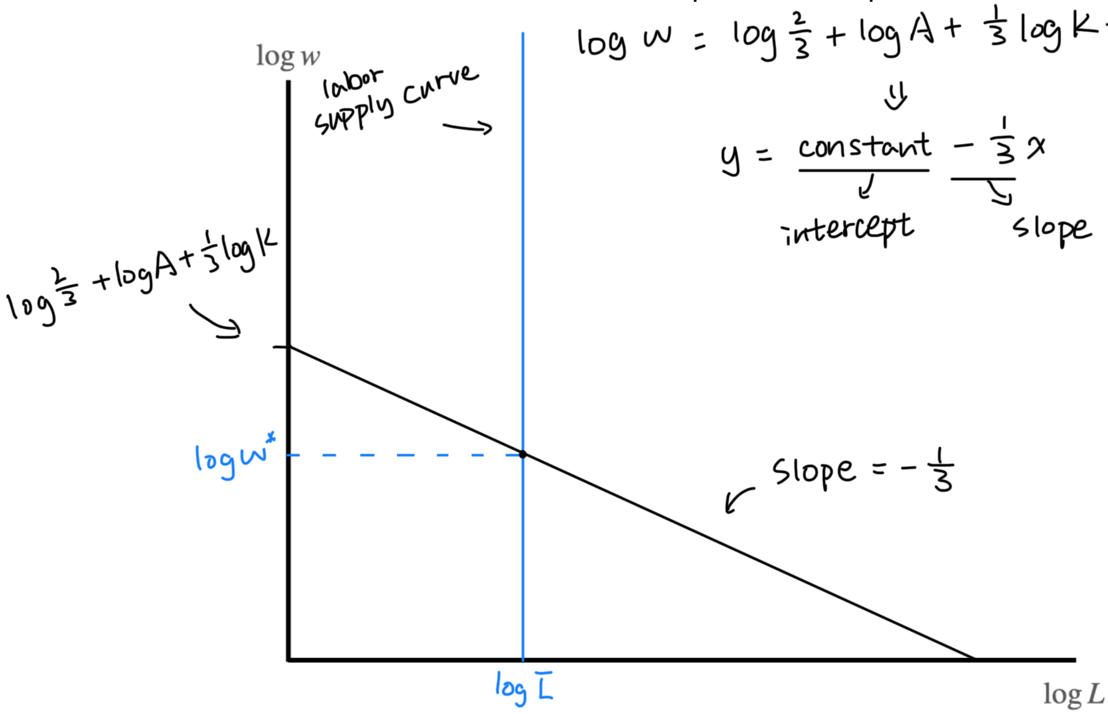
$$\log w = \log \frac{2}{3} + \log A + \frac{1}{3} \log K - \frac{1}{3} \log L \qquad (2.4)$$

(c) [3 points] In the **labor market**, assume technology A and capital K are both fixed. On the axes below, graph the simple relationship that you see in equation (2.4) between the log wage on the vertical axis and $\log L$ on the horizontal axis. Explain what you drew. Report the numerical values of the vertical intercept and the slope.

$$\log w = \log \frac{2}{3} + \log A + \frac{1}{3} \log K - \frac{1}{3} \log L$$

$$\Downarrow$$

$$y = \underbrace{\text{constant}}_{\text{intercept}} \underbrace{- \frac{1}{3}}_{\text{slope}} x$$



A linear relationship between log w and log L. A negative association. Higher labor reduces the marginal product of labor then wage fall.

(d) [2 points] In the picture you just drew, add a vertical **labor supply curve** and show the equilibrium wage and employment in the labor market. Briefly explain below.

With fixed labor supply, no matter how wages change, economy stuck. Then firms adjust the wages they would like to pay depend on the productivity of the workers.

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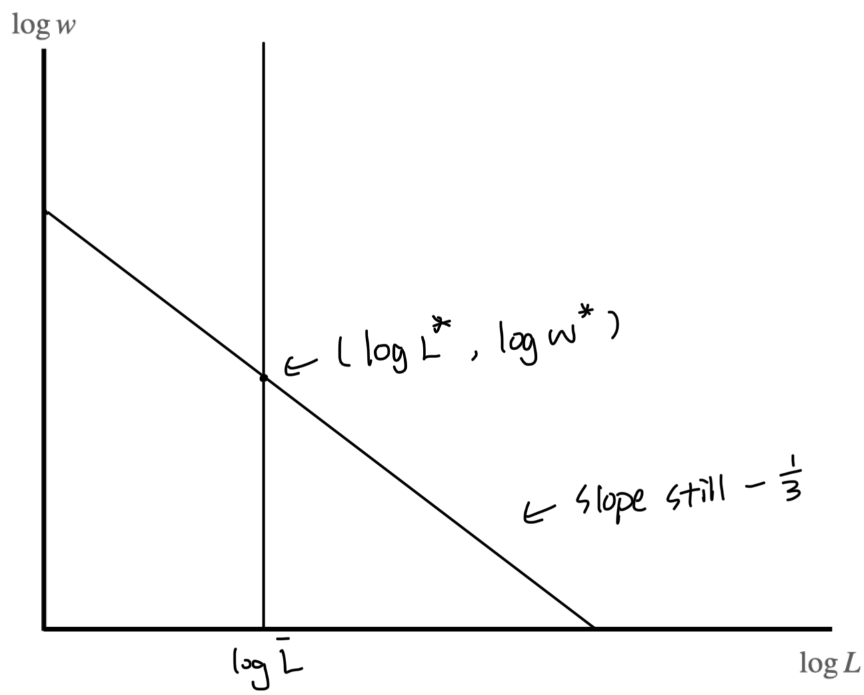
(e) [3 points] Redraw the labor market on the axes below and depict and describe what would happen in the labor market if the amount of **capital** in the economy were to increase, other things equal.



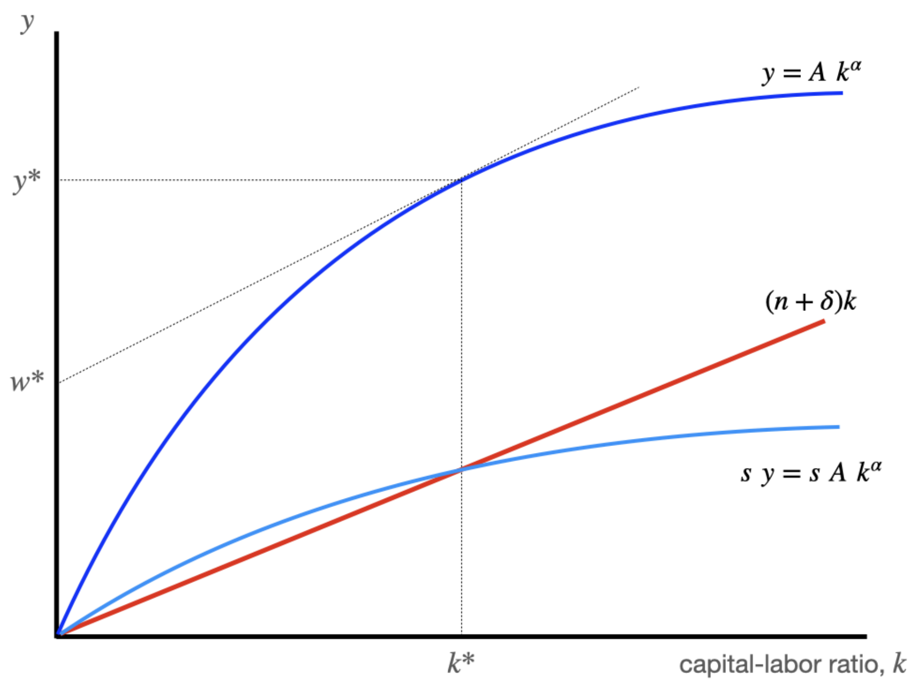
higher K will shift demand up , more money to spend on the employment but the total amount of labor supply fixed . then wage of workers will rise .

3. [16 points total] So how low can we go, Solow? A capital response to immigration?

In the standard neoclassical growth model of Robert Solow without technological growth, consumers supply labor inelastically and save out of income at rate s. There are constant returns to scale in labor and capital, and productivity A is fixed and not changing. Capital depreciates at rate δ, and labor increases at the rate of population growth, n. When output is Cobb-Douglas,

capital per worker is	$k = K/L$	also called the "capital-labor ratio"
output per worker is	$y = A\,k^{\alpha}$	and typically $\alpha = 1/3$
saving per worker is	$s\,y = s\,A\,k^{\alpha}$	
depreciation per worker is	$(n + \delta)k$	

The Solow diagram below shows how capital per worker reaches a *steady state* at k*, where depreciation per worker equals saving per worker: $(n + \delta)k = s\,y$, so capital per worker is unchanging there. The steady state is shown by the vertical dashed line in the diagram below.



(a) [2 points] If the capital per worker is in steady state at k* and not changing over time, what is true about the wage w*, which is shown in the graph on the y-axis?

A and k^* are constants. $\rightarrow$ $y = A\,k^{\alpha}$ fixed to a point that the marginal product of labor is constant, then the wage w^* will also be a constant.

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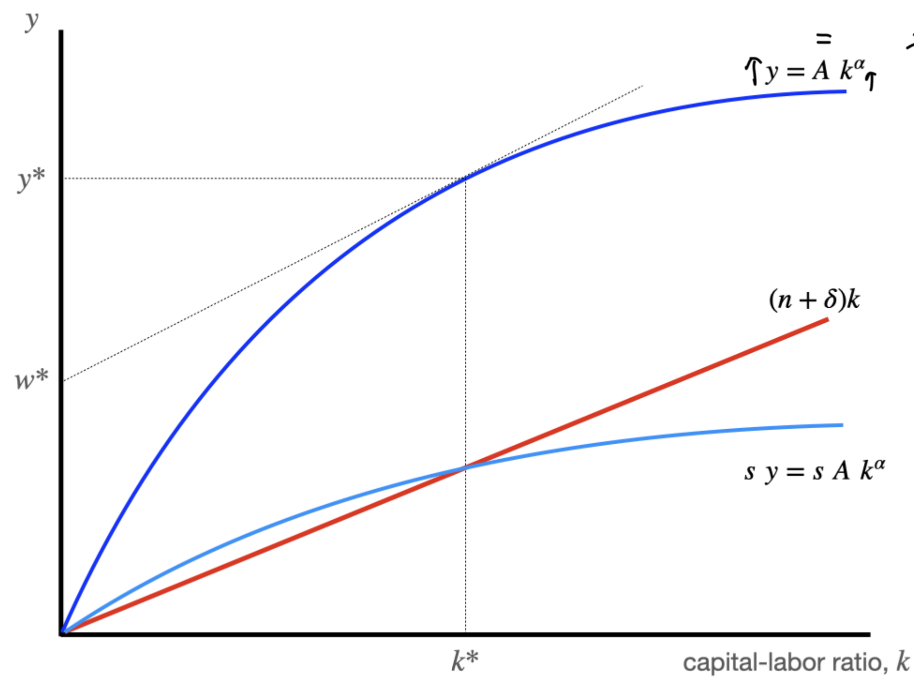
Suppose that the economy is in steady state, and then the government carries out a large removal of immigrant workers, one that increases capital per worker k by about 25%. The policy removes workers but leaves capital K and productivity, A, unchanged.

$k \uparrow = \frac{K =}{L \downarrow}$

(b) [4 points] Illustrate this policy by augmenting the graph from part (a) as needed. Draw any new curves and show jumps and changes in variables. What happens to the wage in the short run? What happens to the wage in the long run? Explain fully. [Hint: do any of the parameters like n, δ, s, A, and α change?]



In short run, the wage w jumps up while removing Labor amount.
In long run, since the Solow stedy state k^* is unchanged, k will larger than k^* thus k will decline over time back to k^*, finally wages just temporarily rise and then falls back.

(c) [2 points] A higher level of capital per worker will produce a higher wage, because workers are more productive. And removing any type of worker will raise capital per worker. But is that change temporary or permanent? Why?

Temporary. Removing workers raises capital per worker and raise the wage. That's true, but the Solow steady state is related to s, A, n, δ and those are unchanged. thus $k > k^*$ then will fall down back to k^* in the future. thus temporory, not short-term.

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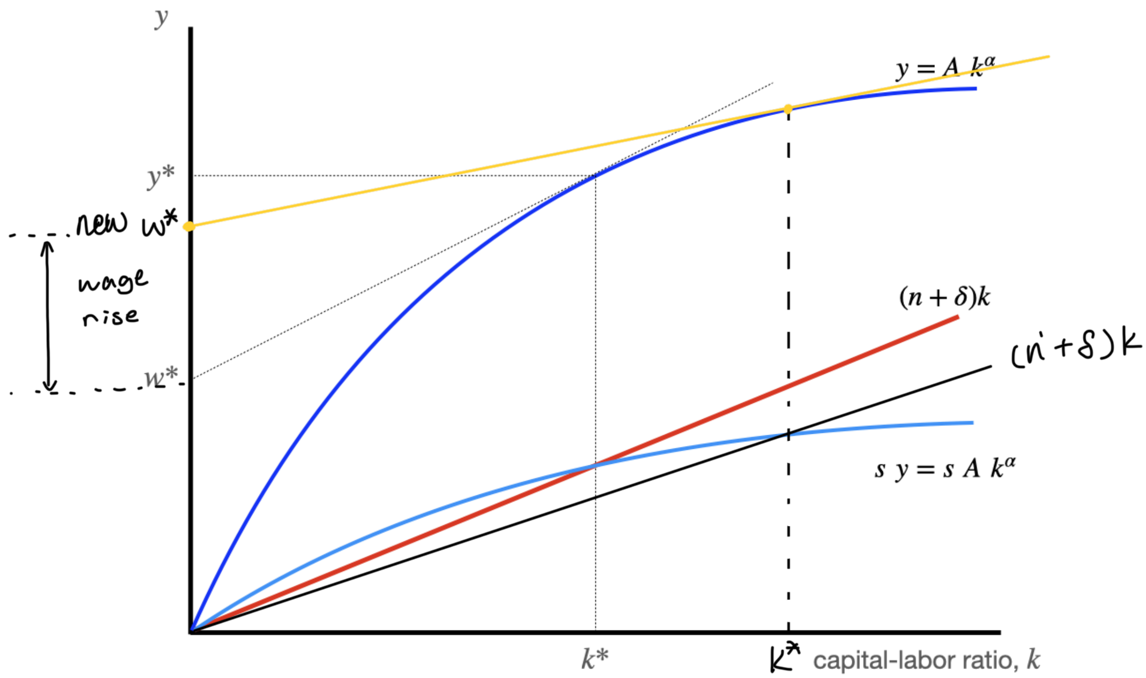
Now suppose that while in steady state, the government instead decides to change **immigration policy**, reducing number of refugee and work visas each year, which reduces the rate of population growth, n, to a new level $n' < n$. (Assume there is no removal policy.)

(d) [4 points] Illustrate this policy by augmenting the graph from part (a) as needed. Draw any new curves and show jumps and changes in variables. What happens to the wage in the short run? What happens to the wage in the long run? Explain fully. [Hint: do any of the parameters like n, δ, s, A, and α change?]



$n' < n \to (n'+\delta)k < (n+\delta)k \to$ smaller slope $\to$ intercept goes up-right.
$\Rightarrow k^*$ goes higher $\Rightarrow$ k will increase to new $k^* \to$ wage permanently higher
short run it doesn't change but long run rise up.

(e) [4 points] What would have happened in the graph in pa rt (d) if immigration enhanced productivity, A, so that reducing immigration reduced the level of productivity? Explain.

$n \downarrow$
$A \downarrow$

Same, n lower and k^* higher, wage rises in long run.
But when lower n causes lower A, lower A causes lower wages
In this situation, change on k^* depends on the impacts of A and n, k^* can be both increase or decrease, then derive to wage, can go up or down.
These are ambiguous due to unstatement of how strong the impacts are.